

CHETAN RAJ VC

CYBER SECURITY AND ETHICAL HACKING PROFESSIONAL

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CAREER OBJECTIVE

To leverage my diverse experience in engineering, management, and cybersecurity to secure a challenging role as a Cybersecurity Engineer, utilizing my skills in ethical hacking, penetration testing, and vulnerability assessment to protect organizations from cyber threats.

SUMMARY

Experienced engineering professional with a background in instrumentation engineering, project management, and team leadership. Proven ability to adapt and thrive in diverse environments, from corporate to startup settings. Possess a strong foundation in Python programming and experience with ethical hacking tools, with completed certifications in ethical hacking and cybersecurity.

WORK EXPERIENCE

Zillionenglish | Manager
June 2012 – March 2024

Managed an online English learning platform based in Kannur

Successfully trained students from all over the world

Led a team of online English language experts.

Prepared candidates for IELTS exams

EDUCATION

National Institute of Business Management
March 2015 – March 2016

MBA in Oil and Gas

Cochin University of Science and Technology
September 2005 - July 2009

B-Tech in Electronics and Instrumentation Engineering

**Galfar Engineering and Contracting, UAE |
Instrumentation Engineer October 2011 – May
2012**

Worked in Abu Dhabi for an instrumentation
contracting company in the oil and gas industry

Reported to the Senior Instrumentation Engineer

Responsibilities included:

- Erection, installation, power-up, and loop checking of field instruments
- Inspection, calibration, commissioning, and troubleshooting of field instruments

CERTIFICATIONS

Ethical Hacking training from Internshala (IIT-Madras approved) - June 2020 to August 2024

Covered Network Security and Penetration Testing with hands-on experience

Cyber Security and Ethical Hacking training from Apponix Technologies - May 2024 to August 2024

Covered Cyber Security Fundamentals, including tools like Nmap, Wireshark, and Nessus

Proficient in Ethical Hacking training from Cadd Centre - May 2024 to July 2024

Hands-on training in Kali Linux and basic ethical hacking concepts

Python programming basics from G-Tec - February 2024 to April 2024

Learned basic programming concepts in Python

TECHNICAL SKILLS

- Encryption
- Social Engineering
- Malwares
- Ethical Hacking Tools
- Risk Assessment
- Vulnerability Assessment
- WAPT(Web Application Penetration Testing)
- Operating Systems: Kali Linux, Windows, Metasploitable
- Familiarity with network protocols and devices

SOFT SKILLS

- Problem-solving and analytical thinking
- Adaptability and time management
- Team leadership and management
- Communication and interpersonal skills

TOOLS

- Nmap
- Wireshark
- Metasploit
- Nessus
- SQL Map

INTERESTS

- Participating in cybersecurity challenges on HackTheBox platform
- Reading books on cybersecurity, technology, and self-improvement
- Playing chess to enhance strategic thinking and problem-solving skills

PROJECT EXPERIENCE

March 2024 -PRESENT

Network Scanning and Enumeration using Nmap

Scope: Conducted a comprehensive network scanning and enumeration on a target system using Nmap to identify open ports, services, vulnerabilities, and weaknesses

Tool(s): Nmap, Kali Linux (attacker machine), Windows 7 (victim machine), VMware (virtualization platform used to set up the hacking lab environment)

Environment:

- Hacking lab setup with Kali Linux as the attacker machine and Windows 7 as the victim machine, both running on VMware
- Firewall temporarily disabled on the victim machine to allow for comprehensive scanning and accurate results

Observation: Scanned the target system using Nmap with the following

command: `sudo nmap -sS 192.168.159.133 -A -sV -F` and Identified open ports and services such as 135/tcp: MSRPC (Microsoft Windows RPC), 139/tcp: NetBIOS-SSN (Microsoft Windows netbios-ssn), 445/tcp: Microsoft-DS (Windows 7 Ultimate 7600 microsoft-ds), 5357/tcp: HTTP (Microsoft HTTPAPI httpd 2.0), 49152/tcp: MSRPC (Microsoft Windows RPC), 49153/tcp: MSRPC (Microsoft Windows RPC), 49154/tcp: MSRPC (Microsoft Windows RPC), 49155/tcp: MSRPC (Microsoft Windows RPC), 49156/tcp: MSRPC (Microsoft Windows RPC)

- **Operating System Detection:** Microsoft Windows 7 SP0 - SP1, Windows Server 2008 SP1, Windows Server 2008 R2, Windows 8, or Windows 8.1 Update 1
- **Device type:** General purpose

Moreover, Checked for vulnerabilities in the NVD database and identified:

- CVE-2020-1472 (MSRPC vulnerability) with a severity of High
- CVE-2020-1350 (NetBIOS-SSN vulnerability) with a severity of Medium
- CVSS score: 8.1 (High)
- Mitre ATT&CK framework: T1190 (Exploit Public-Facing Application), T1110 (Brute Force)

Outcome:

- Discovered potential vulnerabilities in MSRPC and NetBIOS-SSN services due to outdated versions and misconfiguration
- Identified possible attack vectors for exploitation

Mitigation:

- Implemented firewall rules to close open ports and restrict access to vulnerable services
- Updated MSRPC and NetBIOS-SSN services to the latest version, addressing the identified vulnerabilities
- Configured additional security measures, such as intrusion detection and access controls
- Emphasized the importance of regular updates and patch management to prevent similar vulnerabilities, ensuring timely application of security patches and updates to mitigate potential risks

Web Application Security Testing (WAPT): SQL Injection Attack and Vulnerability Assessment on (testphp.vulnweb.com)

Scope: Conduct a comprehensive vulnerability assessment and SQL injection attack on the live website (testphp.vulnweb.com) to identify and exploit potential vulnerabilities, leveraging MITRE ATT&CK techniques (T1190: Exploit Public-Facing Application)

Tool(s): Kali Linux, VMware, SQLmap (version 1.5.4), Firefox

Vulnerability Identification:

- Identified parameter tampering vulnerability in the website's URL, classified as CWE-89: Improper Neutralization of Special Elements used in an SQL Command ('SQL Injection')
- Discovered potential for blind SQL injection attacks, with a CVSS score of 8.5 (High Severity)

Exploitation:

Used SQLmap to exploit the SQL injection vulnerability, utilizing the following commands in KaliLinux terminal:

- sqlmap -u http://testphp.vulnweb.com/listproducts.php?cat=1%27 --dbs
 - sqlmap -u http://testphp.vulnweb.com/listproducts.php?cat=1%27 -D acuart --tables
 - sqlmap -u http://testphp.vulnweb.com/listproducts.php?cat=1%27 -D acuart -T users --columns
 - sqlmap -u http://testphp.vulnweb.com/listproducts.php?cat=1%27 -D acuart -T users -C name,uname,pass,email,cc,phone --dump
- Successfully extracted databases (acuart and information_schema) and tables (8 tables in acuart database)

Post-Exploitation:

- Retrieved sensitive user data, including usernames, passwords, emails, credit card numbers, and phone numbers
- Demonstrated ability to bypass error handling and extract confidential information, highlighting the need for improved security measures

Results and Analysis:

- Successfully compromised the website's security and extracted sensitive data, with a potential impact score of 9.0 (Critical)
- Identified vulnerabilities in error handling and database security, classified as CWE-209: Information Exposure Through Error Messages
- Analyzed the potential impact of the exploited vulnerabilities, referencing MITRE ATT&CK techniques (T1190: Exploit Public-Facing Application)

Risk Assessment and Impact Analysis:

- Identified high-risk vulnerabilities (CVSS score 8.5) that could lead to sensitive data exposure and compromise the website's security
- Assessed the potential impact of the exploited vulnerabilities, including unauthorized access to sensitive data and disruption of business operations
- Determined the likelihood and potential impact of future attacks, highlighting the need for prompt remediation and security measures

Security Recommendations:

- Implement proper input validation and sanitization to prevent SQL injection attacks (CWE-89)
- Use parameterized queries or prepared statements to prevent SQL injection (CWE-91)
- Improve error handling to prevent information disclosure (CWE-209)
- Regularly update and patch the website's software and plugins to prevent exploitation of known vulnerabilities

Languages Known

- English
- Malayalam
- Russian
- Tamil
- Hindi

Language Test Result

IELTS Overall Band Score: 7

Declaration

I hereby declare that the information provided is true and accurate to the best of my knowledge.